

Type of Submission

Assignment for the lecture "E8069 Computational Macroeconomics."

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Declaration of AI use for this assignment:

For the purposes of this assignment, AI tools were used for the following purposes only:

- Question 1, Part B:
 - Coding Debugging
 - Coding Assistance for plot function
- Question 2, Part A:
 - Coding Debugging for Stata
 - Coding assistance via explanation of Stata syntax
- Question 3, Part B:
 - Coding Assistance for Excel
 - Coding Debugging for Excel
 - Supplementary Assistance on Dataset Access
- All other questions: AI use was limited to text review and auxiliary understanding of specific question formulations.

Problem 1: Question 1: The model, pen and paper

A planner with initial capital $k_0 > 0$ chooses consumption and capital to maximize

$$\sum_{t=0}^{\infty} \beta^t u(c_t), \quad u(c) = \frac{c^{1-\gamma} - 1}{1-\gamma}, \quad u(c) = \ln c \text{ if } \gamma = 1, \quad (1)$$

subject to, for every $t \geq 0$,

$$c_t + k_{t+1} = k_t^\alpha + (1 - \delta)k_t, \quad c_t > 0, \quad k_{t+1} \geq 0, \quad (2)$$

with $\alpha \in (0, 1)$, $\beta \in (0, 1)$, $\gamma > 0$, and $\delta \in (0, 1]$.

Throughout, a prime on a variable denotes tomorrow, k' is tomorrow's capital and c' tomorrow's consumption. Derivatives carry the variable as a subscript, $v_k(k) = \partial v / \partial k$ and $u_c(c)$ for marginal utility.

Question 1.1: Recursive formulation and optimality conditions

20 points

- (a) Bring the model into the form of the general sequence problem (SP) from the lecture: identify x_t , the feasible set $\Gamma(x_t)$, and the reward function $F(x_t, x_{t+1})$. Explain in one

or two sentences why k_t is the right choice of x_t , that is, why it summarizes everything about the past that matters for the future, and what would change if the utility function depended on c_{t-1} as well.

- (b) Derive the Bellman equation from the sequence problem. Substitute the budget constraint (2) into the Bellman equation so that k' is the only choice variable and the constraint disappears. Argue that the remaining constraints $c > 0$ and $k' \geq 0$ never bind, so that the optimum is interior.
- (c) Derive the first-order condition for k' and the envelope condition. Derive the envelope condition for the problem at hand, by plugging the policy into the Bellman equation and differentiating with respect to k , rather than quoting the general result. Combine the two conditions into the Euler equation

$$u_c(c) = \beta R(k') u_c(c'), \quad R(k') := \alpha k'^{\alpha-1} + 1 - \delta, \quad (3)$$

where $k' = g(k)$ is the policy function,

$$c = h(k) = k^\alpha + (1 - \delta)k - g(k)$$

the consumption function, and $c' = h(k')$.

- (d) Interpret (3) in words. What is $R(k')$, and why does $1 - \delta$ appear in it?

Question 1.2: A check that needs no closed form

10 points

The Euler equation (3) holds at the exact solution for every k . Whatever a numerical method returns can be plugged into it, and the residual measures how far the candidate is from satisfying optimality. This works in any model with an Euler equation, whatever form it takes, so the check remains available when no closed form exists. To make the residual comparable across states and parameters it is normalized to consumption units. This normalized residual is called the Euler equation error and is written as

$$e(k) := 1 - \frac{u_c^{-1}(\beta R(k') u_c(c'))}{c}, \quad k' = g(k), \quad c = h(k), \quad c' = h(k'). \quad (4)$$

- (a) Show that for CRRA utility (4) simplifies to

$$e(k) = 1 - (\beta R(k'))^{-1/\gamma} \frac{c'}{c}. \quad (5)$$

- (b) On a grid $\mathcal{K} = \{k_1, \dots, k_N\}$ the policy is the index vector $\mathbf{g}$ with $k' = k_{g_i}$ at state k_i , and the consumption function on the grid is the vector $\mathbf{c}$ with

$$c_i = h(k_i) = k_i^\alpha + (1 - \delta)k_i - k_{g_i}.$$

Write down e_i , the Euler equation error at grid point i , in terms of $\mathbf{g}$, $\mathbf{c}$, $\mathcal{K}$, and the parameters.

Question 1.1: Recursive formulation and optimality conditions

Subquestion a: General Sequence Problem

To bring the consumption-saving problem under CRRA utility into the general sequence problem form, we first construct the sequential Lagrangian and substitute in the budget constraint into the utility function to eliminate consumption as a function argument and the constraint itself before iterating the sequential problem forward:

$$\begin{aligned}
\mathcal{L} : & \max_{c_t, k_{t+1}} \frac{c_t^{1-\gamma} - 1}{1-\gamma} + \lambda_t (k_t^\alpha + (1-\delta)k_t - c_t - k_{t+1}) \\
s.t. : & c_t = k_t^\alpha - k_{t+1} + (1-\delta)k_t \\
\Leftrightarrow & \max_{k_{t+1}} \frac{(k_t^\alpha - k_{t+1} + (1-\delta)k_t)^{1-\gamma} - 1}{1-\gamma} \\
\Leftrightarrow & F(k_t, k_{t+1}) = \frac{(k_t^\alpha - k_{t+1} + (1-\delta)k_t)^{1-\gamma} - 1}{1-\gamma} \\
s.t. : & \Gamma(k_t) \in [0, k_t^\alpha + (1-\delta)k_t]; k_0 \text{ given}
\end{aligned} \tag{1}$$

For $\gamma \neq 1$, the reward function above holds. However, for $\gamma = 1$, the CRRA utility function converges to log utility and the reward function is derived as:

$$\begin{aligned}
\Leftrightarrow & F(k_t, k_{t+1}) = \ln(k_t^\alpha - k_{t+1} + (1-\delta)k_t) \\
s.t. : & \Gamma(k_t) \in [0, k_t^\alpha + (1-\delta)k_t]; k_0 \text{ given}
\end{aligned} \tag{2}$$

k_t is sufficient as the sole state variable x_t as it determines both output via the production function $y = k_t^\alpha$, the available capital stock after current depreciation via the law of motion of capital (as implied through the budget constraint), and consumption via the sequential budget constraint itself. Due to the sequential nature of the problem, k_t also represents the realized choice for k_{t+1} in the previous period and subsequently the past consumption and output sequence $\{c_j, y_j\}_{j \leq t}$.

However, if the utility function also depends on past consumption c_{t-1} , it effectively enters the sequential problem as an additional state variable and $x_t = (c_{t-1}, k_t)$. As c_{t-1} depends on k_{t-1} and k_t , we can replace the state variable c_{t-1} with k_{t-1} as well to represent the state variables as $x_t = (k_{t-1}, k_t)$. The latter representation simplifies

Subquestion b: Bellman Equation Representation

To rewrite the general sequence problem into the Bellman equation, we pull out the reward function for the current period from equation (1) and simplify the equation:

$$\begin{aligned}
v^*(k_0) &= \max_{k_{t+1}} \sum_{t=0}^{\infty} \beta^t \left[\frac{(k_t^\alpha - k_{t+1} + (1-\delta)k_t)^{1-\gamma} - 1}{1-\gamma} \right] \\
\Leftrightarrow v^*(k_0) &= \max_{k_{t+1} \in \Gamma(k_0)} \frac{(k_t^\alpha - k_{t+1} + (1-\delta)k_t)^{1-\gamma} - 1}{1-\gamma} \\
&\quad + \beta \max_{k_{t+2}, k_{t+3}, \dots \in \Gamma(k_0)} \sum_{t=0}^{\infty} \beta^t \left[\frac{(k_t^\alpha - k_{t+1} + (1-\delta)k_t)^{1-\gamma} - 1}{1-\gamma} \right] \\
\Leftrightarrow v^*(k_0) &= \max_{k' \in \Gamma(k_0)} \frac{(k^\alpha - k' + (1-\delta)k)^{1-\gamma} - 1}{1-\gamma} + \beta v^*(k')
\end{aligned} \tag{3}$$

To confirm that the constraints $c > 0$ and $k' \geq 0$ hold, we examine the limits of the marginal utility of the underlying CRRA utility function. For $\lim_{c \rightarrow 0^+}$, the limit is derived as

$$\lim_{c \rightarrow 0^+} c^{-\gamma} = \infty$$

for any $\gamma > 0$. This implies, that the agent will always prefer a marginal unit of consumption when the constraint binds, therefore a period with null consumption is never optimal. As the constraint for consumption is never binding, the constraint on future capital $k' \geq 0$ can also never bind. If $k' = 0$, consumption in the next period is consequently zero, which, as seen before, is never optimal. Therefore, the constraint on future capital also never binds and optimum of the consumption-saving problem is always an interior solution, as the only optimal corner solution would be for $k_0 = 0$, which is excluded by assumption on strictly positive consumption $c > 0$.

Subquestion c: Deriving the Euler Equation

We start with the FOC of the Bellman equation w.r.t. k' , which is:

$$\begin{aligned}
FOC_{k'} : & - (k^\alpha - k' + (1-\delta)k)^{-\gamma} + \beta \frac{\partial v^*(k')}{\partial k'} = 0 \\
\Leftrightarrow & (k^\alpha - k' + (1-\delta)k)^{-\gamma} = \beta \frac{\partial v^*(k')}{\partial k'}
\end{aligned} \tag{4}$$

To evaluate the derivative $\frac{\partial v^*(k')}{\partial k'}$, we express $v^*(k')$ as $v^*(g(k))$ where $k' = g(k)$. Therefore the Bellman equation can be expressed as:

$$\begin{aligned}
v^*(k_0) &= \max_{k' \in \Gamma(k_0)} \frac{(k^\alpha - k' + (1-\delta)k)^{1-\gamma}}{1-\gamma} + \beta v^*(k') \\
s.t. \quad & v^*(k') = v^*(g(k)) \\
s.t. \quad & k' = g(k) \\
\Leftrightarrow v^*(k_0) &= \frac{(k^\alpha - g(k) + (1-\delta)k)^{1-\gamma}}{1-\gamma} + \beta v(g(k)) \\
\frac{\partial v^*(k_0)}{\partial k} &= (k^\alpha - g(k) + (1-\delta)k)^{-\gamma} (\alpha k^{\alpha-1} + (1-\delta) - g'(k)) + \beta \underbrace{\left[\frac{\partial v(g(k))}{\partial g(k)} g'(k) \right]}_{= \frac{\partial v(g(k))}{\partial k}}
\end{aligned} \tag{5}$$

Rewriting the equation 5 to isolate $g'(k)$ yields:

$$\begin{aligned}
\frac{\partial v^*(k)}{\partial k} &= (k^\alpha - g(k) + (1 - \delta)k)^{-\gamma} (\alpha k^{\alpha-1} + (1 - \delta) - g'(k)) + \beta \frac{\partial v(g(k))}{\partial g(k)} g'(k) \\
s.t. \ u_c(c) &= (k^\alpha - g(k) + (1 - \delta)k)^{-\gamma} \\
\Leftrightarrow \frac{\partial v^*(k)}{\partial k} &= u_c(c)(\alpha k^{\alpha-1} + (1 - \delta)) - u_c(c)g'(k) + \beta \frac{\partial v(g(k))}{\partial g(k)} g'(k) \\
\Leftrightarrow \frac{\partial v^*(k)}{\partial k} &= u_c(c)(\alpha k^{\alpha-1} + (1 - \delta)) + g'(k) \underbrace{\left(\beta \frac{\partial v(g(k))}{\partial g(k)} - u_c(c) \right)}_{=0 \text{ as per } FOC_{k'} \text{ from equation 4}}
\end{aligned} \tag{6}$$

From equation 4, we know that $\beta \frac{\partial v^*(k')}{\partial k'} = (k^\alpha - k' + (1 - \delta)k)^{-\gamma} = u_c(c)$. This first order condition implies that $\beta \frac{\partial v^*(k')}{\partial k'} - u_c(c) = 0$. Therefore, the entire expression $g'(k)(\beta \frac{\partial v(g(k))}{\partial g(k)} - u_c(c))$ is zero in equilibrium, which is in adherence to the envelope condition and the derivative $\frac{\partial v^*(k)}{\partial k}$ can be expressed as:

$$\frac{\partial v^*(k)}{\partial k} = \frac{\alpha k^{\alpha-1} + (1 - \delta)}{(k^\alpha - g(k) + (1 - \delta)k)^{-\gamma}} \tag{7}$$

Plugging the expression from equation 7 for k' into the FOC from equation 4 finally yields:

$$\frac{1}{(k^\alpha - k' + (1 - \delta)k)^\gamma} = \beta \frac{\alpha (k')^{\alpha-1} + (1 - \delta)}{((k')^\alpha - g(k') + (1 - \delta)k')^\gamma} \tag{8}$$

or in terms of consumption

$$\frac{1}{c^\gamma} = \beta \frac{\alpha (k')^{\alpha-1} + (1 - \delta)}{(c')^\gamma} \tag{9}$$

Both expressions are equivalent to:

$$\begin{aligned}
u_c(c) &= \beta R(k') u_c(c') \\
R(k') &= \alpha k^{\alpha-1} + (1 - \delta) \\
s.t. \ k' &= g(k) \\
c &= h(k) = k^\alpha + (1 - \delta)k - g(k) \\
c' &= h(k')
\end{aligned} \tag{10}$$

Subquestion d: Economic Interpretation of the Euler equation:

The term $R(k')$ represents the marginal return of additional capital investment in form of increasing output and higher future capital stock. The term $1 - \delta$ appears in the net marginal return of capital investment as capital depreciation implies that one unit of capital investment is not equivalent to the change in capital stock.

Question 1.2: A check that needs no closed form

Subquestion a:

$$\begin{aligned}
e(k) &= 1 - \frac{u_c^{-1}(\beta R(k')u_c(c'))}{c} \\
s.t. \ u_c^{-1}(\beta R(k')u_c(c')) &= [(\beta R(k')u_c(c'))]^{-\frac{1}{\gamma}} \\
\Leftrightarrow &= 1 - \frac{[(\beta R(k')u_c(c'))]^{-\frac{1}{\gamma}}}{c} \\
\Leftrightarrow &= 1 - \frac{[(\beta R(k'))]^{-\frac{1}{\gamma}} c'^{-\gamma-\frac{1}{\gamma}}}{c} \\
\Leftrightarrow \ e(k) &= 1 - [\beta R(k')]^{-\frac{1}{\gamma}} \frac{c'}{c}
\end{aligned} \tag{11}$$

Subquestion b:

Start from the simplified Euler equation error and substitute the variables with their equivalent grid positions:

$$\begin{aligned}
e(k) &= 1 - [\beta R(k')]^{-\frac{1}{\gamma}} \frac{c'}{c} \\
s.t. \ k' &= k_{g_i} \in \mathcal{K} = \{k_1, \dots, k_N\} \\
k &= k_i \in \mathcal{K} = \{k_1, \dots, k_N\} \\
c &= c_i = h(k_i) = k_i^\alpha + (1 - \delta)k_i - k_{g_i} \\
c' &= c_{g_i} = h(k_{g_i}) \\
\Leftrightarrow \ e_i(k_i) &= 1 - [\beta R(k_{g_i})]^{-\frac{1}{\gamma}} \frac{c_{g_i}}{c_i} \\
\Leftrightarrow \ e_i(k_i) &= 1 - [\beta(\alpha k_{g_i}^{\alpha-1} + (1 - \delta))]^{-\frac{1}{\gamma}} \frac{c_{g_i}}{c_i}
\end{aligned} \tag{12}$$